

HASAN TURHAN

Türkiye | Open to relocation across Germany

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PROFILE

Software Engineering student with international research experience in AI, computer vision, and multimodal systems, alongside hands-on software development in Python, Java, Kotlin, JavaScript, and SQL. Experienced in building evaluation pipelines, working with vision-language models, and developing Android and data-driven applications. Seeking an Erasmus+ traineeship in Software Engineering or AI/ML in Germany.

EXPERIENCE

University of Beira Interior

AI & Computer Vision Research Trainee

Covilhã, Portugal

Jun. 2026 – Sep. 2026

- Designed and implemented a decomposition agent (classify → read → detect orientation → compute → verify) for VLM-based physical-action prediction, improving movement-prediction accuracy by up to 41 percentage points across evaluated models.
- Benchmarked 16 open-weight vision-language models, including GLM-4.1V, InternVL3.5, Qwen2.5-VL, Qwen3-VL, Ovis2, Molmo, and Pixtral, on a physical-action-prediction task using vLLM on a single 32 GB GPU.
- Validated 2,442 benchmark images and metadata records across real-world (1,272) and synthetic (1,170) splits, checking image-to-metadata mappings, numeric values, intervals, and scale-direction annotations.
- Built Python validation pipelines for Excel and Parquet metadata, automatically detecting inconsistent annotations and flagging ambiguous cases for manual review.

EDUCATION

Malatya Turgut Özal University

B.Sc. in Software Engineering, GPA: 3.34/4.00

Malatya, Türkiye

2023 – Jun. 2027 (Expected)

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Vision, Artificial Intelligence, Machine Learning

RESEARCH OUTPUT

- **Manuscript in preparation:** *From Instrument Reading to Physical-Action Prediction* — Co-author.

PROJECTS

Smart German Vocabulary Trainer | Kotlin, Jetpack Compose, MVVM, DataStore

GitHub

- Built a native Android vocabulary-learning application with interactive flashcards, color-coded German articles, pronunciation, favorites, and user-created vocabulary entries.
- Implemented MVVM architecture, reusable Jetpack Compose components, and persistent local application state using DataStore.

Pharmacy Management System | SQL Server, HTML, CSS, JavaScript

- Designed a SQL Server relational database for medicines, customers, suppliers, sales, and inventory, including CRUD-oriented database operations.
- Built a responsive browser interface with search, stock alerts, daily-sales tracking, input validation, and localStorage-based persistence.

TECHNICAL SKILLS

Programming: Python, Java, Kotlin, JavaScript, SQL, HTML, CSS

AI / ML: PyTorch, OpenCV, vLLM, NumPy, Pandas, Computer Vision, VLM Evaluation, Data Preprocessing

Software / Android: Jetpack Compose, MVVM, DataStore, REST APIs

Data / Tools: Git, GitHub, Linux/WSL2, CUDA, SQL Server, Parquet

LANGUAGES

English: C1 | **German:** B1 | **Turkish:** Native | **Kurdish:** Native